When Configuration Matters More Than Algorithm

An Empirical Evaluation of Lightweight Drift Detectors Under Resource Constraints

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Motivation

- Concept drift: $P(X, y)$ changes over time $\Rightarrow$ models degrade silently
- Existing detector benchmarks assume **unlimited resources**: continuous monitoring, unlimited memory
- Real deployments (edge, IoT, cost-constrained) cannot do that
- Practitioners must choose: **how often** to check, **how much** recent data, **how much** reference history
- **Gap:** no study quantifies how much these choices matter

Research Question

Which matters more for detection performance: the *algorithm*, or *how it is configured*?

Methodology

3 datasets (tabular, binary)

- Credit Default (30k)
- Telco Churn (7k)
- Predictive Maint. (10k)

3 detectors

- KS test, PSI, Wasserstein

3 drift types

- Sudden, gradual, recurring

Constraint grid (3×3×3)

- Monitoring frequency
- Window size
- Reference size

8,100 experimental runs + 405 model-comparison runs

Non-parametric stats: Kruskal-Wallis, Mann-Whitney U (Bonferroni-corrected), rank-biserial r, Spearman's ρ

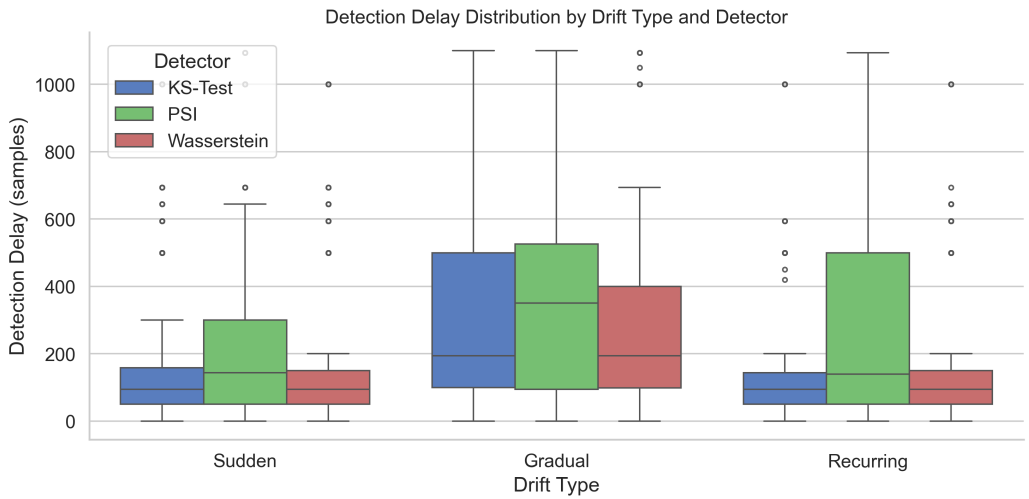
RQ1 & RQ2: Detector and Drift-Type Effects

Detector choice (small effects, $|r| \leq 0.18$)

- Wasserstein: lowest delay & missed detections
- PSI: cheapest compute, but slowest & noisiest
- KS vs. Wasserstein: **statistically indistinguishable** ($p = 0.186$)

Drift type (largest effect in the study, $r \approx 0.30$)

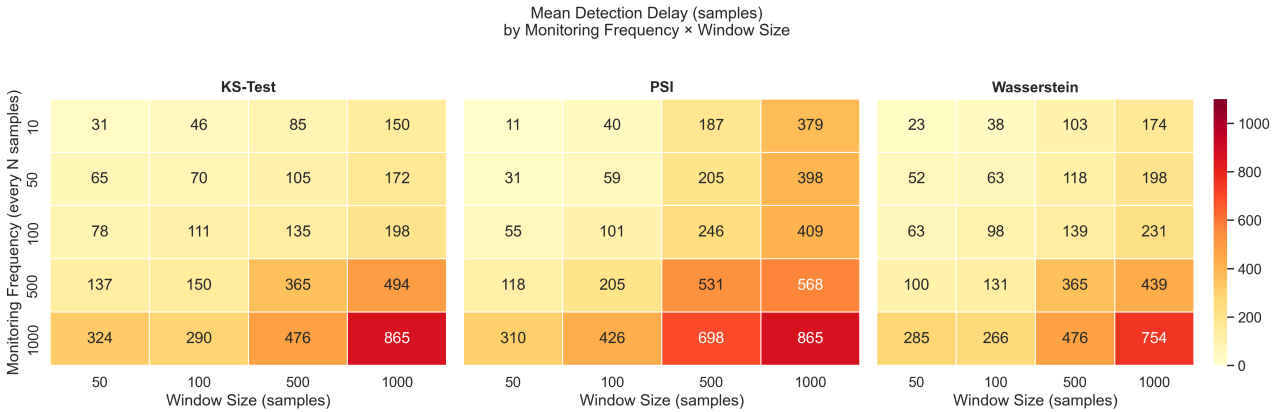
- Gradual drift is significantly harder to detect
- Sudden $\approx$ Recurring ($p = 1.00$): same detection mechanism



RQ3: Constraint Configuration Dominates

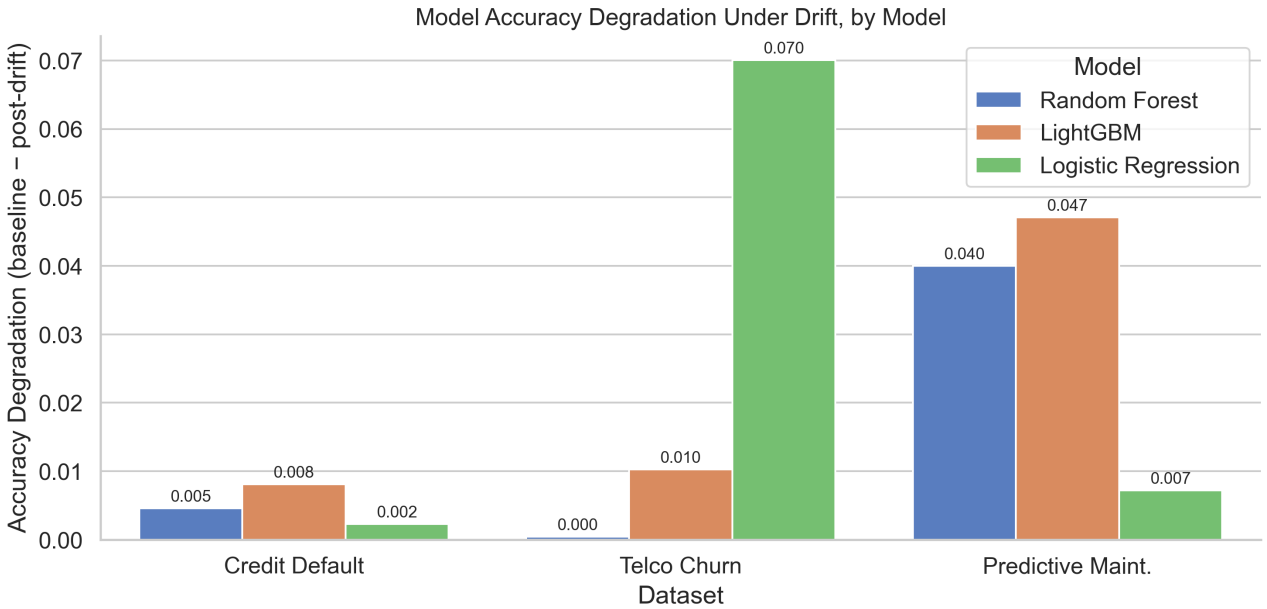
37-fold range in detection delay
(22.0 windows best case → 827.6 windows worst case)

- Window size: $\rho = 0.543$ (strongest single effect)
- Monitoring frequency: $\rho = 0.487$
- Reference size: a **threshold**, not a scale (500 samples ≈ unlimited)



RQ4: Model Robustness Under Drift

- **Random Forest** degrades least under drift
- vs. LightGBM: $r = -0.383$ (medium) vs. Log. Reg.: $r = -0.185$ (small)
- Mechanism: bagging + feature-subset randomisation **decorrelates** trees, so a shift in one feature does not propagate to the whole ensemble
- LightGBM's sequential boosting **propagates** shift-induced error across rounds



Priority Hierarchy for Practitioners

1. **Monitoring regime first**: frequency & window size (37× impact)
2. **Drift type awareness second**: gradual drift needs tighter configuration
3. **Detector/model choice third**: secondary; choose on cost or robustness, not raw performance

Reframing

The field asks "which algorithm is best?" This study shows: **ask how it is configured first.**

Limitations & Next Steps

Limitations

- Synthetic/injected drift, not organic real-world drift
- Univariate detection only (per-feature, not correlation structure)
- 3 datasets, tabular binary classification only

For discussion

- Testing scale-sensitivity via distribution-preserving synthetic expansion (not duplication)
- Candidate venues: IEEE COMPSAC SRS, SAICSIT 2026
- Chapter 5/6 merge decision